Mohammad Elsharqawy

LLM Systems & AI Infrastructure

Mohammad_Elsharqawy@f-eng.tanta.edu.eg · mohammadelsharqawyegy@gmail.com

Egypt · GitHub · LinkedIn · Portfolio

Education

Tanta University — M.Sc. in Computer Engineering Nov. 2024 – Present
GPA: 3.78/4.00 Tanta, Egypt

Tanta University — B.Sc. in Computer Engineering and Automatic Control Oct. 2018 – Jun. 2023
Graduated *Excellent with Honors* (the highest possible grade); ranked 4th of 150+ students.

Research Experience

Research Intern — System Reliability & Agentic SRE May 2026 – Aug. 2026
University of Illinois Urbana-Champaign (UIUC); advisor: Prof. Tianyin Xu *Remote*

- Developed reproducible Kubernetes failure scenarios to evaluate LLM-based SRE agents on diagnosis and remediation.
- Studied incident-derived benchmark fidelity and agent limitations, including cross-namespace reasoning and configuration-to-symptom correlation.

Research Assistant — Robustness under Distribution Shift Oct. 2025 – Present
Tanta University; advisor: Prof. Mohamed T. Faheem Saidahmd *Tanta, Egypt*

- Investigating generalizable diabetic retinopathy grading across imaging hardware, acquisition protocols, and patient populations.

Research Software Contributions

libCacheSim — High-Performance Cache Library and Simulator 2026
MultiQueue (MQ) eviction; project led by Prof. Juncheng Yang (Harvard) Merged PR #318

- Implemented MQ eviction in C (∼550 LOC); integrated queue management, ghost buffering, registration, and regression tests.
- Verified LRU equivalence with one queue, passed the eviction test suite, and evaluated miss ratios against seven baselines across workloads and cache sizes.

SREGym — Benchmark for Agentic Site Reliability Engineering (UIUC) 2026
High-fidelity Kubernetes fault scenarios PR #774 · PR #791

- Reproduced GKE pod-IP exhaustion using Calico IPAM, with deterministic setup and cleanup.
- Built a configuration-triggered latent-bug scenario: HTTP 500 responses despite healthy-running pods, with fault activation and mitigation checks.

Selected Projects

LLM Inference Engineering — Systems from First Principles GitHub repository

- Investigated LLM inference internals, including attention, KV-cache behavior, model-weight movement, and compute-bound prefill versus memory-bound decoding.
- Developed PyTorch experiments to analyze attention FLOPs, memory traffic, arithmetic intensity, batching, and sequence-length scaling using roofline-based reasoning.
- Benchmarked GPU workloads: decode throughput increased from ∼0.2 TFLOPS (batch 1) to 6.1 TFLOPS (batch 32), while long-sequence prefill reached ∼23 TFLOPS.

Teaching Experience

Teaching Assistant — Tanta University Sep. 2024 – Present

- Led labs, instruction, and assessment in Operating Systems, Data Structures & Algorithms, Artificial Intelligence, Machine Learning, OOP, and programming.
- Coached 11 teams for the 2023 ECPC Qualifications, including Neutrinos (4th place; subsequently competed at ACPC).

Additional Experience

AI, Data Science & Cloud Engineering Trainee — Digital Egypt Builders Initiative Jan. – Sep. 2024
Selected for a national program covering AI, data science, and cloud engineering.

Software Engineering Intern — Ithra Services Ltd. Jul. – Sep. 2022
Built REST APIs with MongoDB/JWT; containerized services and automated AWS EC2 deployments with CI/CD and Ansible.

Honors & Awards

- Top Students Financial Award, Tanta University (five consecutive years, 2019–2023).
- ICPC: 2nd, Tanta Collegiate Contest (2021); 8th, ECPC Qualifications (2022); ECPC contestant (2021–2022).

Technical Skills

Programming: Python, C/C++, Go, Java, JavaScript, Bash
Systems & Infrastructure: Linux, Kubernetes, Docker, Calico CNI, AWS, Terraform, Ansible, GitHub Actions
ML & AI Systems: PyTorch, Transformers, LLM inference, ONNX Runtime, MLflow, RAG, LoRA
Observability & Data: Prometheus, Grafana, Elasticsearch, Jaeger, MySQL, MongoDB